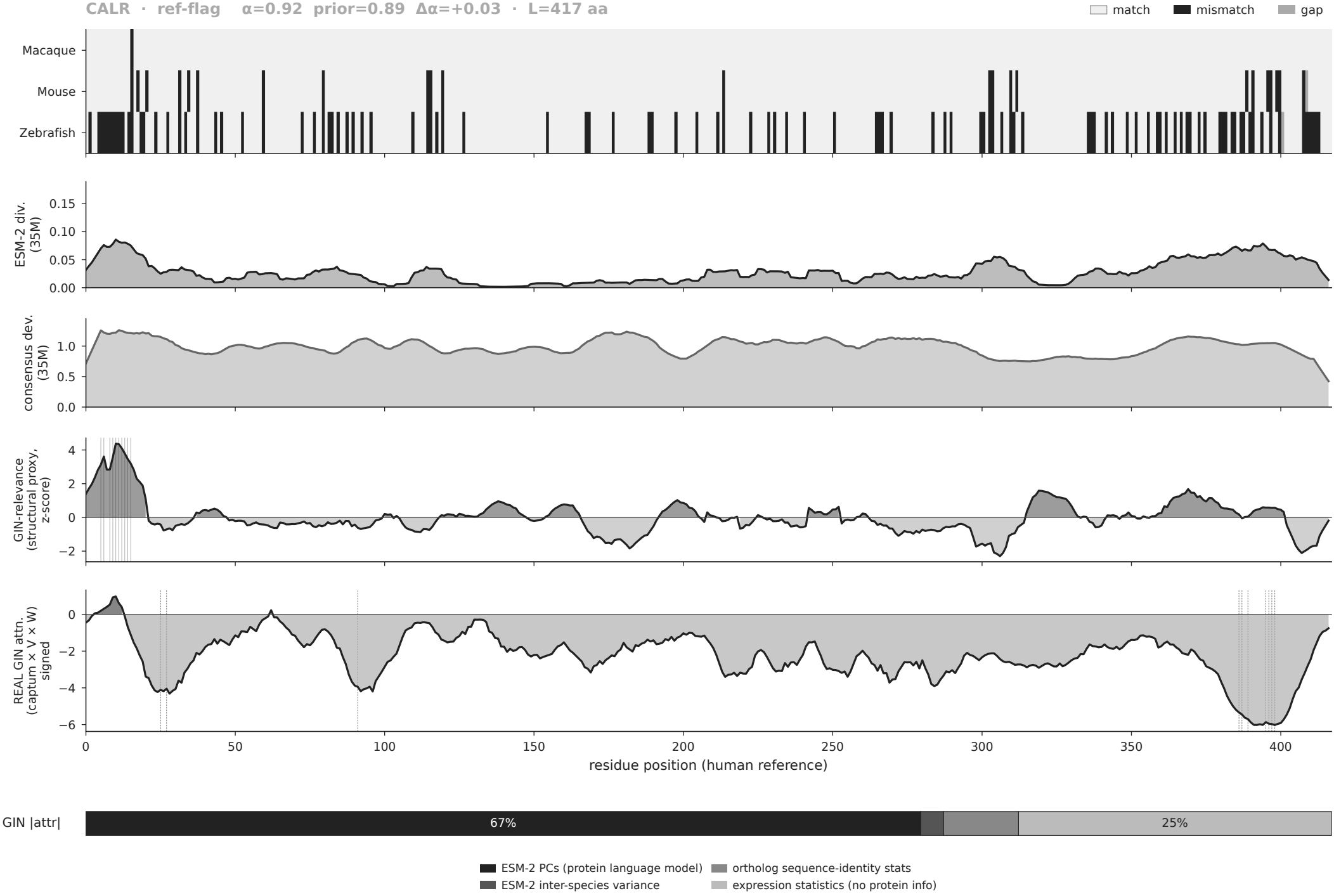
CALR · ref-flag α=0.92 prior=0.89 Δα=+0.03 · L=417 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.15
0.10
0.05
0.00
consensus dev. (35M)
1.0
0.5
0.0
GIN-relevance (structural proxy, z-score)
4
2
0
−2
REAL GIN attn. (captum × V × W) signed
0
−2
−4
−6
0
50
100
150
200
250
300
350
400
residue position (human reference)
GIN |attr|
67%
25%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)